

Instructions: Read the following passage. Key computer science vocabulary has been highlighted in blue. If you do not understand a word, look at the definition below, or in a dictionary. Read two or three times. When you are ready, listen to Baidenmann read, then take the quiz.

Jason's Very Cool Computer

My computer is very cool. I use it to study, play games, watch movies, and do my 3D modeling and graphic design work. The computer is inside a tall tower case with clear glass on the side. I can see the parts inside. There are colorful RGB lights inside, and they slowly change color. The tower sits on my desk next to my monitor.

Inside the tower, the most important part is the **motherboard**. All other parts connect to it. I have the latest motherboard, and it supports the newest **PCIe 5.0** and **DDR5** technology. It has many **slots** and **ports** for hardware. It also has a fast **chipset** that helps all the parts talk to each other.

In the middle of the motherboard is the **CPU**. The CPU is the brain of the computer. My computer uses a very powerful CPU with many **cores** and **threads**. These cores allow the computer to do many tasks at the same time. When I edit graphical elements for work, or play a game for fun, the CPU works very fast. It sits in a **socket** on the motherboard, and over it there is a large **liquid cooling** system. This **cooler** keeps the CPU **temperature** low when the computer works hard.

Next to the CPU, there are four sticks of **RAM**. The RAM is the working memory of the computer. My computer has 32GB of **DDR5** RAM with a very high **clock speed**. Programs move data in and out of the RAM very quickly. This helps games load faster and helps large programs run smoothly.

Another important part is the **GPU**. It is connected to the **PCIe slot** on the motherboard. I have a large, powerful GPU to do my design work and play **graphics intensive AAA games**. The GPU is also used for internet browsing, HDR movies, 3D rendering, and **AI acceleration**. It has many **CUDA cores**, a lot of **VRAM**, and advanced **ray tracing** hardware. When I play the latest games, I can turn the graphics options to maximum.

For storage, my computer uses several **NVMe SSD** drives. These drives connect through the **M.2 interface** on the motherboard. The SSD has very fast **read speed** and **write speed**. Because of this, the operating system starts in just a few seconds. Load times on programs and games are very short. Most programs start instantly.

At the bottom of the tower is the **power supply unit (PSU)**. The PSU gives electricity to all the components. It is a high-wattage **modular PSU**, so the cables are neat and organized. Good **power management** keeps the system stable.

The tower also has many **case fans** for **airflow**. The fans pull cool air into the case and push hot air out. Together with the **liquid cooling radiator**, the whole system stays cool, even when I am running intensive programs.

I have some great peripherals too. My main display is a large OLED **4K monitor** with a high **refresh rate**. The colors are bright and the black color is very deep. When the **GPU** sends **frames** to the monitor, the picture looks smooth and clear at 120 **frames per second (FPS)**.

In front of the monitor is my [mechanical keyboard](#). On a mechanical keyboard, each key has a [mechanical switch](#). The keys are big and make a sound when you press them. It is easy to type quickly. The keyboard also has programmable [macros](#), which help when I code or play games.

Near the monitor I have two [stereo speakers](#), a high precision [gaming mouse](#), and a [USB hub](#). Sometimes I use a high-quality [microphone](#) and [webcam](#) for online meetings. All these devices connect through fast [USB-C ports](#) on the computer.

I built this desktop myself, and I love using it. When new components come out, I will upgrade it. How about you? What internal and external components does your computer have?

Key Computer Science Vocabulary

Instructions: You do not need to memorize all of this vocabulary. Just do your best to understand the passage.

AAA (Triple-A)

AAA games are very large video games made by a large team with lots of money at big game companies.

AI (Artificial Intelligence)

Artificial Intelligence is computer technology that allows machines to perform tasks that usually require human thinking, such as learning or recognizing patterns.

Airflow

Airflow is the movement of air through a computer case to remove heat and keep the hardware cool.

Case Fan

A case fan is a fan inside a computer case that moves air to cool the internal components.

Chipset

A chipset is a group of electronic parts on the motherboard that control communication between the CPU, memory, storage, and other hardware.

Clock Speed

Clock speed is the rate at which a processor or memory works, usually measured in gigahertz (GHz).

Compile

Compile means to convert programming code written by humans into machine code that a computer can run.

Component

A component is any physical hardware part inside or connected to a computer system.

CPU (Central Processing Unit)

The Central Processing Unit is the main processor that performs calculations and controls most operations in a computer.

Core (CPU Core)

A core is an individual processing unit inside a CPU that can run instructions and tasks.

CUDA (Compute Unified Device Architecture)

CUDA is a technology from NVIDIA that allows a GPU to perform general computing tasks, not only graphics processing.

DDR5 (Double Data Rate 5)

DDR5 is a modern type of RAM that can transfer data very quickly between memory and the CPU.

Desktop Computer

A desktop computer is a personal computer designed to stay in one place, usually on or under a desk. A laptop computer is a portable computer.

Display

A display is a screen device that shows images, text, and video from a computer.

DIMM (Dual Inline Memory Module)

the DIMM is the hardware that processes the computer's run-time memory.

FPS (Frames Per Second)

Frames per second is the number of images a computer sends to a display every second in video or games.

GPU (Graphics Processing Unit)

The Graphics Processing Unit is a specialized processor that creates images, video, and 3D graphics for a computer.

HDR (High Dynamic Range)

High Dynamic Range is a display technology that shows brighter colors and greater contrast between light and dark areas.

Interface

An interface is a connection point or standard that allows different hardware devices to communicate.

Mechanical Keyboard

A mechanical keyboard is a keyboard where each key uses a physical mechanical switch for better typing feedback.

Liquid Cooling System

A liquid cooling system is a cooling method that uses liquid to carry heat away from computer components.

M.2 (M-dot-2 Interface)

M.2 is a small connector interface used to attach high-speed storage devices such as NVMe SSDs to a motherboard.

Macro (Programmable Macro)

A macro is a programmed sequence of commands that can run automatically when a key or button is pressed.

Microphone

A microphone is an input device that captures sound and sends it to a computer.

Monitor

A monitor is a display screen that shows visual output from a computer.

Motherboard

A motherboard is the main circuit board that connects and allows communication between all computer components.

Mouse (Gaming Mouse)

A mouse is a pointing device that allows a user to control the cursor and interact with items on a computer screen.

NVMe (Non-Volatile Memory Express)

NVMe is a fast communication protocol that allows SSD storage devices to transfer data quickly through PCIe.

OLED (Organic Light-Emitting Diode)

OLED is a display technology where each pixel produces its own light, creating very bright colors and deep blacks.

OS (Operating System)

An operating system is the main software that manages hardware and allows users to run programs.

PCIe (Peripheral Component Interconnect Express)

PCIe is a high-speed interface used to connect devices such as GPUs and SSDs to the motherboard.

Peripheral

A peripheral is an external device connected to a computer, such as a keyboard, mouse, or printer.

Port, socket, slot

These three words have the same meaning. A port, socket, or slot is a place in the computer where cables or devices can be plugged in. For example, USB ports, PCIe slots, or the CPU socket.

PSU (Power Supply Unit)

The Power Supply Unit converts electricity from a wall outlet into the correct power for computer components.

Radiator

A radiator is a cooling device that releases or 'radiates' heat from cooling systems into the air.

RAM (Random Access Memory)

Random Access Memory is temporary working memory where a computer stores data that programs are currently using.

Ray Tracing

Ray tracing is a graphics technique that simulates how light moves to create very realistic images.

Read and Write Speed

Read and write speed is the rate at which a storage device can transfer and save data.

Rendering (3D Rendering)

Rendering is the process of generating an image from a 3D model using computer calculations.

RGB (Red, Green, Blue)

RGB refers to a lighting system that mixes red, green, and blue light to create many colors.

SSD (Solid State Drive)

A Solid State Drive is a storage device that uses flash memory to store data quickly without moving parts.

Stereo Speakers

Stereo speakers are audio output devices that play sound from a computer through two channels.

Storage

Storage is hardware that keeps data and programs permanently on a computer.

Thread (CPU Thread)

A thread is a sequence of instructions that a CPU core can process independently.

Tower Case

A tower case is the vertical enclosure that holds the internal hardware components of a desktop computer.

Upgrade

An upgrade is the process of replacing or adding hardware to improve a computer's performance.

USB (Universal Serial Bus)

Universal Serial Bus is a common standard used to connect peripherals and transfer data between devices.

USB-C (Universal Serial Bus Type-C)

USB-C is a modern USB connector type that supports fast data transfer, power delivery, and device connections.

USB Hub

A USB hub is a device that allows multiple USB devices to connect to a single USB port.

VRAM (Video Random Access Memory)

VRAM is special memory on a GPU used to store image and graphics data.

Webcam

A webcam is a small camera connected to a computer that records or streams video.